CSCE 2501
Fundamentals of Database Systems

Project - Resale Properties

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Introduction

In this project, you will be building a database application for the real estate properties in Egypt that are offered for Resale, that allows users to get information about these properties in the country, the different developments, locations, etc.

The main site that provide such information is AqarExit website (https://aqarexit.com/en/opportunities?show=8000). However, the website has limited abilities for filtering or doing some higher level aggregations on the market size and state. Hence, your application will be used to provide more flexibility on that front.

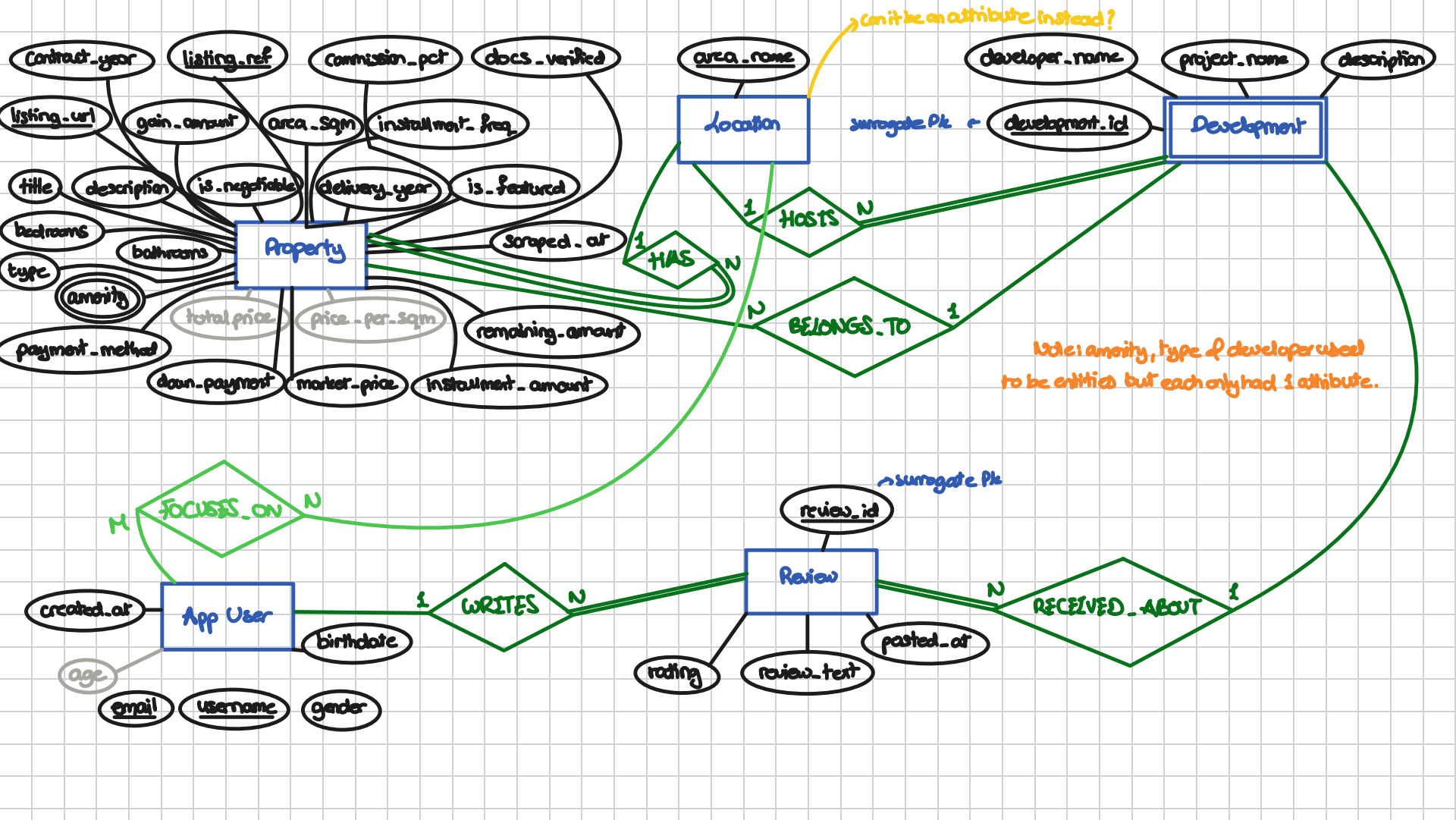
Your task is to build a database system backend and an application that provide the resale market analytics and features described next.

Milestone I: Data Requirements

The database system you design should be based on the information provided on each property, its location, developer, type, and the other information provided on the property page (e.g. sample)

Your database design should store all the information about the resale property, including but not limited to: area, location, development, type (apartment, villa, etc.), number of bedrooms and bathrooms, list of amenities, description, payment method, down payment, gain over current value, etc. Many of the properties on the website are located in a development project, for such cases the development project needs to be recorded, including its name, location, developer, etc.

The system should also allow users to register on the system using their email addresses, and pick a username, gender, age, birthdate, and their main areas of focus. After registering, the users can add their reviews for any development project, providing their rating (1-10) and a textual review.


Contract_year
listing_ref
commission_pct
docs_verified
area_name
can it be an attribute instead?
developer_name
project_name
description
listing_url
gain_amount
area_sqm
installment_freq
surrogate PK
development_id
Development
title
description
is_negotiable
delivery_year
is_featured
Location
bedrooms
bathrooms
Property
scraped_at
1
HOSTS
N
type
amenity
1
HAS
N
total price
price_per_sqm
N
BELONGS_TO
1
remaining_amount
payment_method
down_payment
market_price
installment_amount
Note: amenity, type & developer used to be entities but each only had 1 attribute.
surrogate PK
review_id
M
FOCUSES_ON
N
Review
created_at
App User
1
WRITES
N
N
RECEIVED_ABOUT
1
birthdate
posted_at
age
rating
review_text
email
username
gender

Property(listing_ref, listing_url, contract_year, commission_pct, docs_verified, gain_amount, is_negotiable, area_sqm, installment_freq, delivery_year, is_featured, title, description, bedrooms, bathrooms, payment_method, down_payment, market_price, installment_amount, scraped_at, type, remaining_amount, area_name, development_id).

Location(area_name)

Development(development_id, project_name, description, developer_name, area_name).

PropertyAmenity(listing_ref, amenity)

Review(review_id, rating, review_text, posted_at, development_id, email).

AppUser(email, username, gender, birthdate, created_at)

UserFocusesLocation(email, area_name)

Assumptions:

- Every record is a resale listing from AqarExit, so one entity (PROPERTY) covers all units (every unit is a resale unit)
- Attributes come from three sources: the assignment text, the AqarExit listing cards, and what the Milestone III queries need.
- Most listing fields are nullable, because a listing may omit them.
- Location is a single level.
- Each development has 1 location only.
- Names on the English pages appear in Arabic (types and areas), and developer names vary in casing. I assume the crawler normalizes them before they go into the lookup tables.
- A development has exactly one location and at most one developer.
- A listing belongs to at most one development, and some listings may have none.
- Each listing has exactly one type.
- Amenities are plain named features, with no quantity or detail per unit.
- All amounts are in EGP, stored as DECIMAL, and areas are in square meters.
- down_payment is the card's "Cash required now" and remaining_amount is "Remaining with developer"
- total_price = down_payment + remaining_amount. On some sample cards from the site, gain = market price − total price − 1.25% × total price matches the displayed gain exactly.
- market_price is the owner's own figure. The site says it is not an Aqar Exit valuation.
- gain_amount and commission_pct are stored as the site shows them, not recomputed.
- installment_frequency is free text.
- Delivery is stored as a year only.
- "Age of contract year" in Milestone III can be computed at query time query time (= current year - contract_year).
- Flags (such as is_featured, docs_verified, etc.) default to false.
- Amounts can't be negative, area must be positive, and years must be between 1990 and 2100.
- Registration needs email, username, gender, birthdate and focus areas. The spec doesn't mention passwords, so none are stored.
- Email and username are unique, and gender is limited to Male, Female and Other (the site forces users to create unique usernames).
- age is derived from birthdate.
- "Main areas of focus" means a set of locations, so each user can pick several.
- Reviews are about development projects, not individual listings, as stated.
- A user can review a given project only once.
- Ratings are integers from 1 to 10, and review text is optional.

```
DROP DATABASE IF EXISTS properties;

CREATE DATABASE properties
    CHARACTER SET utf8mb4
    COLLATE utf8mb4_unicode_ci;      -- utf8mb4: site data contains Arabic text
USE properties;

CREATE TABLE location (
    area_name  VARCHAR(100) NOT NULL,
    CONSTRAINT pk_location PRIMARY KEY (area_name)
) ENGINE=InnoDB;


CREATE TABLE app_user (
    email       VARCHAR(255) NOT NULL,
    username    VARCHAR(50)  NOT NULL,
    gender      ENUM('Male','Female','Other') NOT NULL,
    birthdate   DATE         NOT NULL,
    created_at  TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_app_user PRIMARY KEY (email),
    CONSTRAINT uq_app_user_username UNIQUE (username),
    CONSTRAINT chk_user_email CHECK (email LIKE '%_@_%._%')
) ENGINE=InnoDB;

-- Age is derived from birthdate (CURDATE() cannot be used in a generated
-- column), so it is exposed through a view instead of being stored.
CREATE VIEW v_user AS
SELECT email, username, gender, birthdate,
       TIMESTAMPDIFF(YEAR, birthdate, CURDATE()) AS age,
       created_at
FROM app_user;
```

```
CREATE TABLE development (
    development_id  INT          NOT NULL AUTO_INCREMENT,
    project_name    VARCHAR(150) NOT NULL,
    description     TEXT         NULL,
    developer_name  VARCHAR(150) NULL,      -- NULL if developer not listed
    area_name       VARCHAR(100) NOT NULL,
    CONSTRAINT pk_development PRIMARY KEY (development_id),
    CONSTRAINT uq_development UNIQUE (project_name, area_name),
    CONSTRAINT fk_development_location
        FOREIGN KEY (area_name) REFERENCES location (area_name)
        ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;

CREATE INDEX idx_development_developer ON development (developer_name);
```

```
CREATE TABLE property (
    listing_ref        VARCHAR(30)   NOT NULL,   -- AqarExit id, e.g. U-16437
    listing_url        VARCHAR(500)  NULL,
    title              VARCHAR(255)  NULL,
    description        TEXT          NULL,
    type               VARCHAR(50)   NOT NULL,   -- Apartment, Villa, Townhouse, ...
    area_name          VARCHAR(100)  NOT NULL,
    development_id     INT           NULL,       -- NULL if not in a development
    area_sqm           DECIMAL(10,2) NULL,
    bedrooms           TINYINT UNSIGNED NULL,
    bathrooms          TINYINT UNSIGNED NULL,
    payment_method     VARCHAR(50)   NULL,
    contract_year      SMALLINT      NULL,
    delivery_year      SMALLINT      NULL,
    down_payment       DECIMAL(15,2) NULL,       -- "Cash required now"
    remaining_amount   DECIMAL(15,2) NULL,       -- "Remaining with
developer"
    installment_amount DECIMAL(15,2) NULL,
    installment_freq   VARCHAR(20)   NULL,       -- Quarterly / Semi-annual / ...
    market_price       DECIMAL(15,2) NULL,
    commission_pct     DECIMAL(5,2)  NULL,
    gain_amount        DECIMAL(15,2) NULL,
    is_negotiable      BOOLEAN       NOT NULL DEFAULT FALSE,
    docs_verified      BOOLEAN       NOT NULL DEFAULT FALSE,
    is_featured        BOOLEAN       NOT NULL DEFAULT FALSE,
    scraped_at         TIMESTAMP     NOT NULL DEFAULT
CURRENT_TIMESTAMP,
    CONSTRAINT pk_property PRIMARY KEY (listing_ref),
    CONSTRAINT fk_property_location
        FOREIGN KEY (area_name) REFERENCES location (area_name)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_property_development
        FOREIGN KEY (development_id) REFERENCES development
(development_id)
ON UPDATE CASCADE ON DELETE SET NULL,
    CONSTRAINT chk_property_area     CHECK (area_sqm IS NULL OR
area_sqm > 0),
    CONSTRAINT chk_property_contract CHECK (contract_year IS NULL OR
contract_year BETWEEN 1990 AND 2100),
    CONSTRAINT chk_property_delivery CHECK (delivery_year IS NULL OR
delivery_year BETWEEN 1990 AND 2100),
    CONSTRAINT chk_property_commis   CHECK (commission_pct IS NULL OR
commission_pct BETWEEN 0 AND 100),
    CONSTRAINT chk_property_amounts  CHECK (
        (down_payment       IS NULL OR down_payment       >= 0) AND
        (installment_amount IS NULL OR installment_amount >= 0) AND
        (market_price       IS NULL OR market_price       >= 0))
) ENGINE=InnoDB;

CREATE INDEX idx_property_area_type ON property (area_name, type);
CREATE INDEX idx_property_development ON property (development_id);
CREATE INDEX idx_property_contract ON property (contract_year);


CREATE TABLE property_amenity (
    listing_ref  VARCHAR(30)  NOT NULL,
    amenity      VARCHAR(100) NOT NULL,
    CONSTRAINT pk_property_amenity PRIMARY KEY (listing_ref, amenity),
    CONSTRAINT fk_pa_property
        FOREIGN KEY (listing_ref) REFERENCES property (listing_ref)
        ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE=InnoDB;

CREATE INDEX idx_pa_amenity ON property_amenity (amenity);
```

```
CREATE TABLE review (
    review_id       INT          NOT NULL AUTO_INCREMENT,
    rating          TINYINT UNSIGNED NOT NULL,
    review_text     TEXT         NULL,
    posted_at       TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    development_id  INT          NOT NULL,
    email           VARCHAR(255) NOT NULL,
    CONSTRAINT pk_review PRIMARY KEY (review_id),
    CONSTRAINT uq_review_app_user_dev UNIQUE (email, development_id),  --
one review per app_user per project
    CONSTRAINT chk_review_rating CHECK (rating BETWEEN 1 AND 10),
    CONSTRAINT fk_review_development
        FOREIGN KEY (development_id) REFERENCES development
(development_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_review_app_user
        FOREIGN KEY (email) REFERENCES app_user (email)
        ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE=InnoDB;

CREATE INDEX idx_review_development ON review (development_id);


CREATE TABLE user_focuses_location (
    email      VARCHAR(255) NOT NULL,
    area_name  VARCHAR(100) NOT NULL,
    CONSTRAINT pk_user_focuses_location PRIMARY KEY (email, area_name),
    CONSTRAINT fk_ufl_app_user
        FOREIGN KEY (email) REFERENCES app_user (email)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_ufl_location
        FOREIGN KEY (area_name) REFERENCES location (area_name)
        ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE=InnoDB;
```

```
-- ------------------------------------------------------------------
-- Verification
-- ------------------------------------------------------------------
SHOW TABLES;
DESCRIBE location;
DESCRIBE development;
DESCRIBE property;
DESCRIBE property_amenity;
DESCRIBE app_user;
DESCRIBE review;
DESCRIBE user_focuses_location;
```